WP2. Alliance for Artificial Intelligence in Higher Education – Setup SMARTIE

T2.2. Analysis related to adoption AI in educational process

D.2.2.3. Survey among SMARTIE and other universities

2024-1-RO01-KA220-HED-000244949

Contents

1. Foreword

The rapid development of artificial intelligence has begun to transform the way knowledge is created, delivered, and assessed across the education sector. In higher education in particular, AI-based tools are increasingly influencing teaching practices, learning environments, academic support services, and the overall student experience. From intelligent tutoring systems and adaptive learning platforms to generative AI applications that support content creation, research, and feedback, the integration of AI into the learning process is no longer a future prospect, but a present and growing reality.

2. Introduction

In this context, surveying higher education institutions is of critical importance. Universities and other tertiary education providers play a central role in shaping how AI is adopted, governed, and embedded in pedagogical practice. Understanding their level of readiness, perceptions, needs, concerns, and existing experiences is essential for building an evidence-based picture of the current state of AI adoption in learning. Such surveys make it possible to identify not only the opportunities associated with AI—such as personalization, efficiency, accessibility, and innovation—but also the challenges related to ethics, academic integrity, digital competences, infrastructure, staff preparedness, and institutional policy.

A structured survey among higher education institutions provides valuable insight into how universities interpret the role of AI in teaching and learning, how academic staff and students engage with these technologies, and what forms of support are needed to ensure responsible and effective implementation. It also contributes to a comparative understanding across institutions and countries, helping stakeholders detect common trends, gaps, and good practices. In this way, survey results can inform strategic planning, policy development, curriculum modernization, staff training, and future collaborative initiatives.

Moreover, the importance of such an investigation extends beyond institutional management. At a broader level, it supports the development of more inclusive, resilient, and forward-looking education systems that are capable of responding to technological change while preserving the core values of higher education: quality, equity, critical thinking, academic freedom, and human-centered learning.

This survey was therefore conceived as an essential step in understanding the evolving relationship between artificial intelligence and higher education. By collecting the views and experiences of higher education institutions, it aims to contribute to a more informed dialogue on the pedagogical, organizational, and strategic implications of AI adoption in the learning process, and to support the development of recommendations for its meaningful and responsible integration.

3. Methodology of survey

The survey among universities was implemented through an online data collection methodology designed to ensure accessibility, consistency of responses, efficient data management, and straightforward export of results for subsequent statistical analysis. Since the questionnaire had already been developed prior to the technical implementation stage, the methodological focus was placed on the digital construction of the survey instrument, online distribution to respondents, secure collection of responses, and conversion of the collected data into a structured spreadsheet database. For these purposes, Paperform was selected as the online survey platform, as it provides a document-style form editor, submission management, email-based response workflows, conditional logic, and options for exporting or integrating response data with spreadsheet tools.

The first methodological stage consisted of transposing the already finalized questionnaire into the Paperform environment. In practical terms, this meant creating a new online form and introducing each survey item into the editor according to the original questionnaire structure. Paperform allows the researcher to insert questions directly in the form editor by selecting the appropriate field type for each item, such as text input, email field, or multiple-choice question. The platform documentation indicates that questions are added from the editor by choosing the desired field type and then configuring the question title, description, and specific behavior. It also supports a broad set of field types, which is useful when a university survey includes demographic items, closed questions, scaled answers, or open comments.

Once the questions were entered, the next step was the configuration of survey logic and completion rules. In a survey addressed to universities, it is often necessary to maintain coherence between questions and reduce the burden on respondents. For example, certain questions may only be relevant to respondents from particular institutional categories, departments, or functions. Paperform includes conditional logic that can show or hide specific questions, sections, or pages depending on previous answers. This capability is methodologically important because it improves response quality, limits irrelevant answers, and makes the online questionnaire more user-friendly. In addition, required questions can be configured so that essential items must be answered before submission, while optional questions can remain open for broader qualitative feedback.

A further methodological step involved the configuration of respondent identification and communication fields. In surveys among universities, it is often necessary to know the institution, country, respondent profile, or contact email, either for validation, follow-up, or disaggregation of results. Paperform supports dedicated email questions and allows those fields to be reused in automated workflows after submission. This is relevant when the project team wishes to send an acknowledgment message to respondents, provide a copy of their answers, or confirm participation in the study. The platform documentation specifies that respondent emails can be captured through an email-type question and then used for automated messages triggered after successful form submission.

After the construction of the form, the methodology proceeded to the publication and distribution stage. Once the survey instrument was finalized in the editor, it had to be published so it could be accessed online by university respondents. Paperform's workflow supports sharing the form after publication and collecting responses remotely through a web link. This method is particularly appropriate for higher education institutions distributed across different regions or countries, because it removes the logistical barriers associated with paper questionnaires or face-to-face data collection. The survey link has been distributed by email to institutional representatives, university staff, academic managers, teachers, researchers, or other target groups defined in the sampling plan. In this way, the same standardized questionnaire is delivered to all respondents, ensuring comparability of the collected data.

Regarding how data have been sent and collected from respondents, the methodological flow was straightforward. Each respondent received the survey link and accessed the questionnaire through a browser. After filling in the form, the respondent submitted the completed questionnaire electronically. At that point, the platform recorded the submission in its submissions area, where the research team reviewed all received answers during the analysis stage. The platform stored response data at form level and make it accessible from the dashboard or directly from the form editor through the results/submissions section. The online platform also allowed the assignment of a unique submission identifier automatically, which supported traceability and data management during later analysis. This mechanism ensured that each response can be distinguished as a separate case in the dataset.

An important methodological advantage was that the online platform supported post-submission communication and transmission of data summaries. According to the platform documentation, once a form was submitted, the platform automatically sent a submission summary by email to the form owner, the respondent, or another designated recipient.

In the context of a university survey, this option has been used in several ways:

- to notify ULISBOA partner that a new institutional response has been received
- to provide the respondent with confirmation that the questionnaire was successfully submitted
- to forward submission details to responsible work package leaders.

This approach offered transparency and improved administrative control over the data collection process. The methodology also included provisions for monitoring completion and preserving incomplete data when relevant. Online surveys frequently face the problem of respondent dropout before final submission. The platform documented a feature for viewing partial submissions on selected plans, as well as an automatic save-and-resume-later option on the same device. From a methodological standpoint, these functions were useful because they reduced data loss in longer questionnaires and helped the research team to understand where respondents abandoned the survey. In this study, involving universities, where some respondents interrupted completion because of workload or administrative duties, this feature contributed to improved completion rates and better management of non-response. After data collection, the final methodological component concerned the creation of a spreadsheet database for analysis. The platform provided a native option to download all results in a single CSV file, which has been opened directly in Microsoft Excel and saved as an Excel workbook for the creation of an XLS/XLSX database. In addition, the online platform offered a direct Google Sheets integration, accessible from After Submission → Integrations & Webhooks, where each new form submission has been automatically added as a new row in a selected spreadsheet. During this integration, the researcher maps survey answered to spreadsheet columns, thereby creating a continuously updated tabular database suitable for later statistical analysis, filtering, coding, or visualization. While the official export described by the surveying platform was CSV file, the combination of CSV export and Google Sheets integration provided a practical route for creating an Excel-compatible database of responses. An example, screenshot from consolidated Excel file is presented in Figure 1.

	Which country is your institution located in?	What is your institution's profile?	How many students
1	ES	Medical school	More than 10,000
2	ES	University	More than 10,000
3	IT	University	1,000-5,000
4	RO	University	More than 10,000
5	RO	University	More than 10,000
6	AT	University of applied sciences	1,000-5,000
7	SI	University	More than 10,000
8	IT	University	More than 10,000
9	PT	University	1,000-5,000
10	PT	University	5,001-10,000
11	AT	University of applied sciences	5,001-10,000
12	ES	University	More than 10,000
13	ES	University	1,000-5,000

Figure 1 Structured database with answers

From an operational perspective, the procedure for database generation was therefore as follows:

- first, the research team accessed the form's submissions dashboard;
- second, all responses have been exported in bulk as a CSV file, or/and sent automatically to Google Sheets through the direct integration;
- third, the resulting table was checked for consistency of columns and row structure;
- fourth, the file was opened in Excel and saved in Excel format for coding and statistical processing. This step was essential because it transformed raw online responses into a structured research database, where each row represented one respondent or institution and each column corresponded to one survey variable. Such structuring was necessary for descriptive statistics, cross-tabulations, comparative institutional analysis, and graphical reporting.

4. Conclusion

In conclusion, the survey methodology among universities relied on a fully digital workflow in which an already developed questionnaire was implemented in Paperform, configured with appropriate question types and logic, distributed online through a standardized survey link, and completed by respondents via electronic submission. The responses were stored in Paperform's submissions environment, could trigger automated emails or summaries, and were subsequently exported or integrated into spreadsheet software for database creation and analysis. This approach ensured standardization of the instrument, ease of access for geographically dispersed university participants, efficient collection of data, and compatibility with Excel-based analytical procedures. Overall, the use of Paperform provided a practical and methodologically robust framework for administering the university survey and producing a reliable dataset for subsequent interpretation and reporting.